

Christian pioneers: $^{87}\text{Sr}/^{86}\text{Sr}$ isotope analysis at Kata Gård an early Christian site in Varnhem, Sweden

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Background

- Kata Gård in Varnhem (Fig.1) offers an opportunity to study mobility during the Christianization period (9th–11th c. AD)
- Socially stratified cemetery with ~3000 burials
- Crypt was dated to the 10th c. AD, making it one of the earliest Christian churches in Sweden (Fig. 2)
- Diverse geology (approx. 0.70 to >0.74) (Fig. 3)



Fig. 1 Site location in reference to other early Christian sites (Blue) and modern cities (Red)



Fig. 2 Church crypt

Methods and Materials

- Molar enamel from 25 individuals (12 ♂, 9 ♀, 8 subadults) were sampled
- LA-MC-ICP-MS measured $^{87}\text{Sr}/^{86}\text{Sr}$ (Fig. 3)
- Local ranges defined by 5 km and 30 km (Fig. 4)

Fig. 3 Multiple ablation lines yield high-resolution mobility profiles

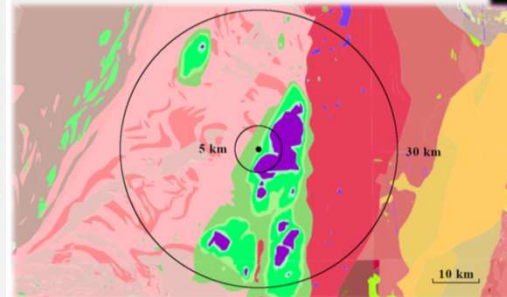


Fig. 4 Geological variation 5-30 km and beyond of Varnhem (black dot)

Results

- Adult ablated lines were mostly nonlocal (86.6%) at 5 km and local (60.8%) at 30 km (Fig. 5)
- Subadult ablated lines were mostly local - 85.3% were local within 5 km and 100.0% within 30 km (Fig. 5)
- Variations in mobility profiles called for additional categories so 5 Mobility categories were determined (see Fig. 6)
 - Nearly 1/2 of the individuals were non-local and originated beyond 5km but only 20% were non-local within 30 km (Figure 6)
- Early Christian site comparison (Fig.1)
 - Sigtuna: Varnhem had more non-locals and more genetic diversity despite no commercial center
 - Alvastra: Large cemetery = Magnet for Christian burial or affiliation

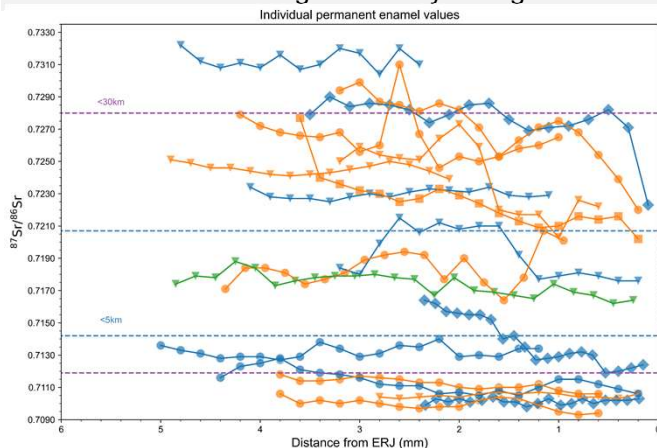
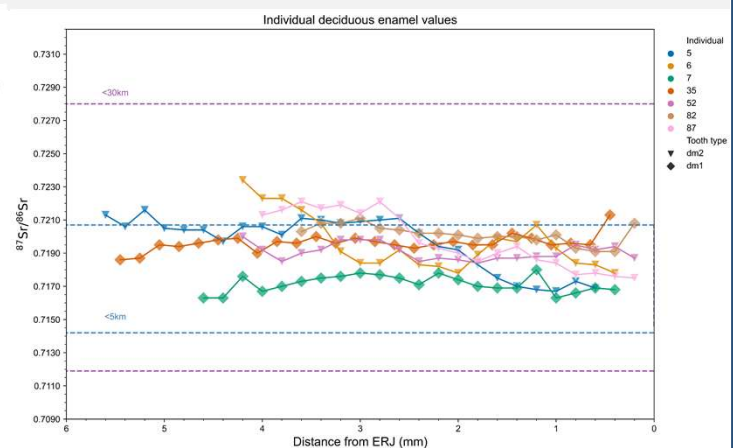


Fig. 5 $^{87}\text{Sr}/^{86}\text{Sr}$ values from deciduous and permanent molars plotted against the distance from the enamel-root junction (ERJ)



Conclusion

- High adult mobility, contrasted with low subadult mobility, suggests that Varnhem attracted incoming families
- Evidence indicates that Varnhem functioned as a regional Christian hub, drawing individuals from within 30 km and beyond
- These patterns highlight Varnhem's central role in the social and religious transformation of the region

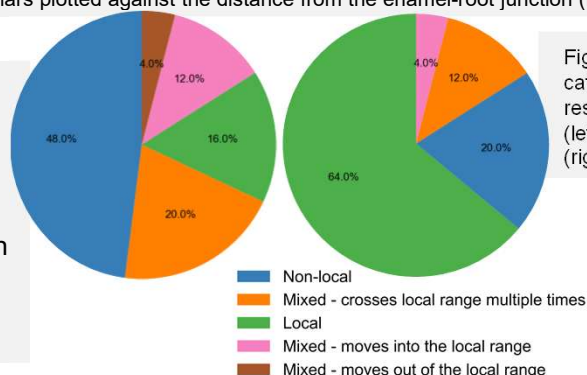


Fig. 6 Mobility categories with results within 5 km (left) and 30 km (right) range